VIT-AP UNIVERSITY, ANDHRA PRADESH

Lab Sheet 6: MongoDB Basic commands

Branch/ Class: B.Tech/M.Tech **Date:26-02-26**
Faculty Name: Prof. S.Gopikrishnan **School:** SCOPE
Student name: Sivagoni Chaitanya Sai **Reg. no.: 23BCE9376**

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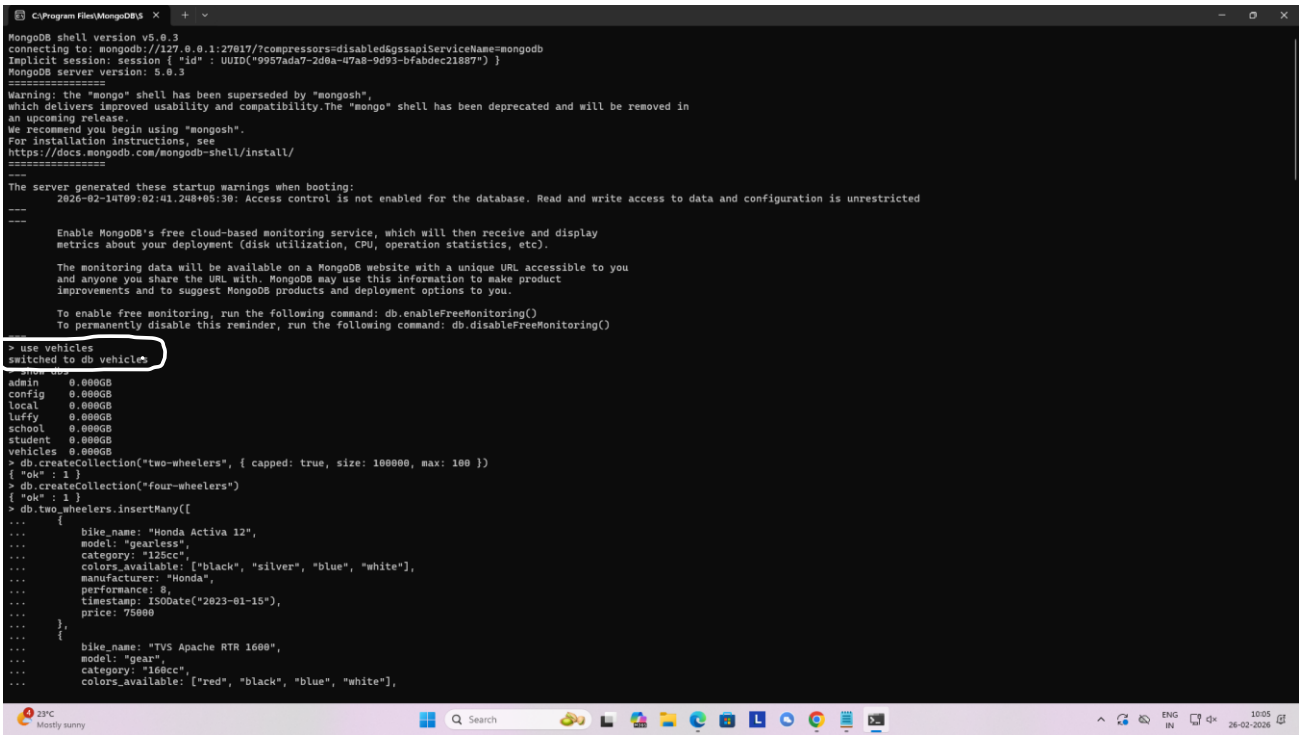
1. Use MongoDB to implement the following DB operations

   1. Create a database called 'vehicles' and *write* a MongoDB query to select database as "vehicles".



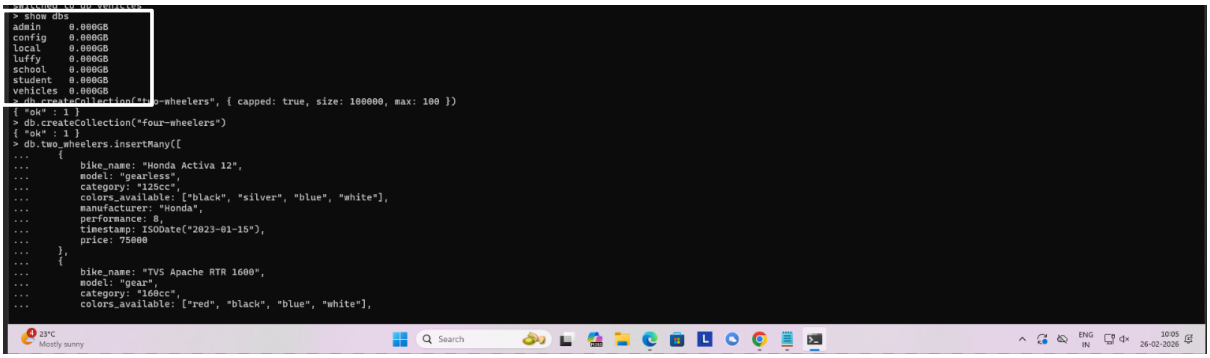
```
C:\Program Files\MongoDB\S
MongoDB shell version v5.0.3
connecting to: mongodb://127.0.0.1:27017/?compressors=disabled&gssapiServiceName=mongodb
Implicit session: session { "id" : UUID("9957ada7-2d0a-47a8-9d93-bfabdec21887") }
MongoDB server version: 5.0.3
================
Warning: the "mongo" shell has been superseded by "mongosh",
which delivers improved usability and compatibility.The "mongo" shell has been deprecated and will be removed in
an upcoming release.
We recommend you begin using "mongosh".
For installation instructions, see
https://docs.mongodb.com/mongodb-shell/install/
================
---
The server generated these startup warnings when booting:
        2026-02-14T09:02:41.248+05:30: Access control is not enabled for the database. Read and write access to data and configuration is unrestricted
---
---
        Enable MongoDB's free cloud-based monitoring service, which will then receive and display
        metrics about your deployment (disk utilization, CPU, operation statistics, etc).

        The monitoring data will be available on a MongoDB website with a unique URL accessible to you
        and anyone you share the URL with. MongoDB may use this information to make product
        improvements and to suggest MongoDB products and deployment options to you.

        To enable free monitoring, run the following command: db.enableFreeMonitoring()
        To permanently disable this reminder, run the following command: db.disableFreeMonitoring()

> use vehicles
switched to db vehicles
admin     0.000GB
config    0.000GB
local     0.000GB
luffy     0.000GB
school    0.000GB
student   0.000GB
vehicles  0.000GB
> db.createCollection("two-wheelers", { capped: true, size: 100000, max: 100 })
{ "ok" : 1 }
> db.createCollection("four-wheelers")
{ "ok" : 1 }
> db.two_wheelers.insertMany([
...     {
...         bike_name: "Honda Activa 12",
...         model: "gearless",
...         category: "125cc",
...         colors_available: ["black", "silver", "blue", "white"],
...         manufacturer: "Honda",
...         performance: 8,
...         timestamp: ISODate("2023-01-15"),
...         price: 75000
...     },
...     {
...         bike_name: "TVS Apache RTR 1600",
...         model: "gear",
...         category: "160cc",
...         colors_available: ["red", "black", "blue", "white"],
```

   2. Write a MongoDB query to display all the databases.



```
> show dbs
admin     0.000GB
config    0.000GB
local     0.000GB
luffy     0.000GB
school    0.000GB
student   0.000GB
vehicles  0.000GB
> db.createCollection("two-wheelers", { capped: true, size: 100000, max: 100 })
{ "ok" : 1 }
> db.createCollection("four-wheelers")
{ "ok" : 1 }
> db.two_wheelers.insertMany([
...     {
...         bike_name: "Honda Activa 12",
...         model: "gearless",
...         category: "125cc",
...         colors_available: ["black", "silver", "blue", "white"],
...         manufacturer: "Honda",
...         performance: 8,
...         timestamp: ISODate("2023-01-15"),
...         price: 75000
...     },
...     {
...         bike_name: "TVS Apache RTR 1600",
...         model: "gear",
...         category: "160cc",
...         colors_available: ["red", "black", "blue", "white"],
```

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Student name: Sivagoni Chaitanya Sai **Reg. no.: 23BCE9376**

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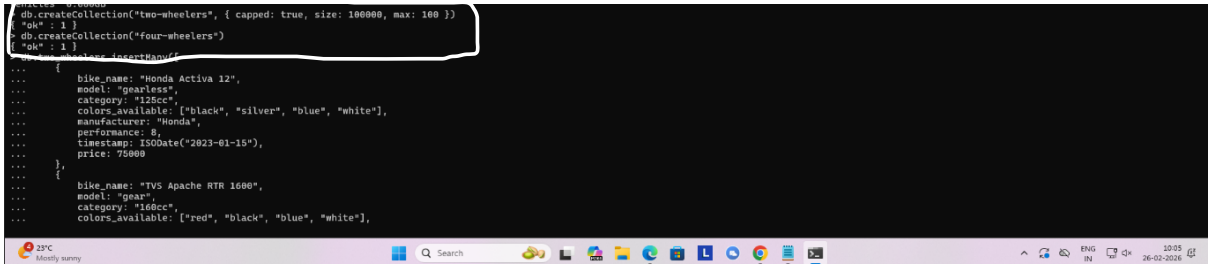
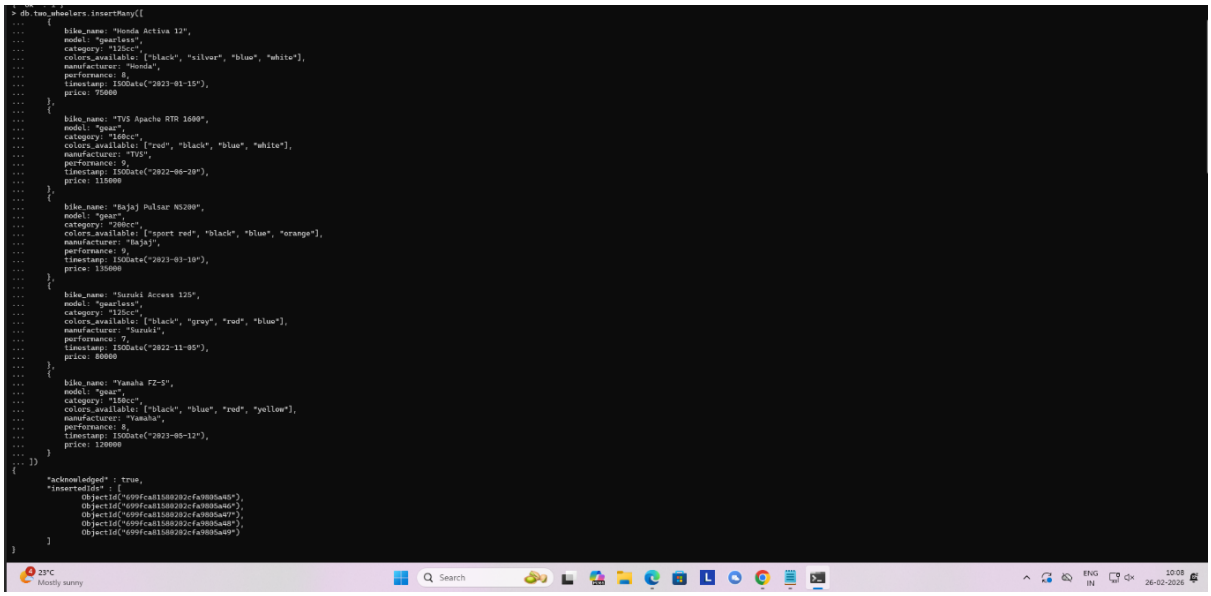
3. Create a collection called 'two_wheelers'. (use capping) and Create a collection called 'four_wheelers'.



4. Add 5 two-wheeler details to the collection named 'two_wheelers'. Each document consists of following fields as bike_name, model (gear or gearless), category (100cc, 125cc, 150cc, 200cc), colors_available (red, black, blue, sport red etc) as array, manufacturer, performance (out of 10), timestamp (date and year release) and price.



5. Add 5 four-wheeler details to the collection named 'four_wheelers'. Each document consists of following fields as vehicle_name, model (commercial or own), category (car, lorry, bus, mini truck, heavy truck, containers), variants (vxi, zxi, petrol, diesel etc) as array, manufacturer, performance (out of 10), timestamp (date and year release) and price.

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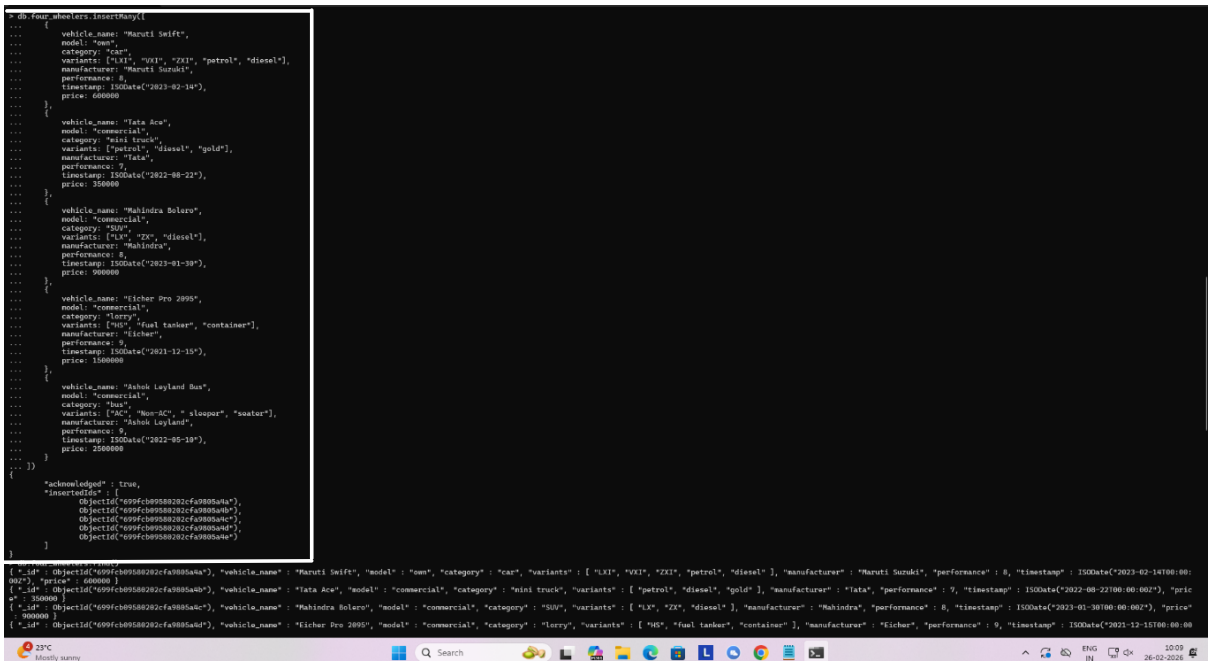
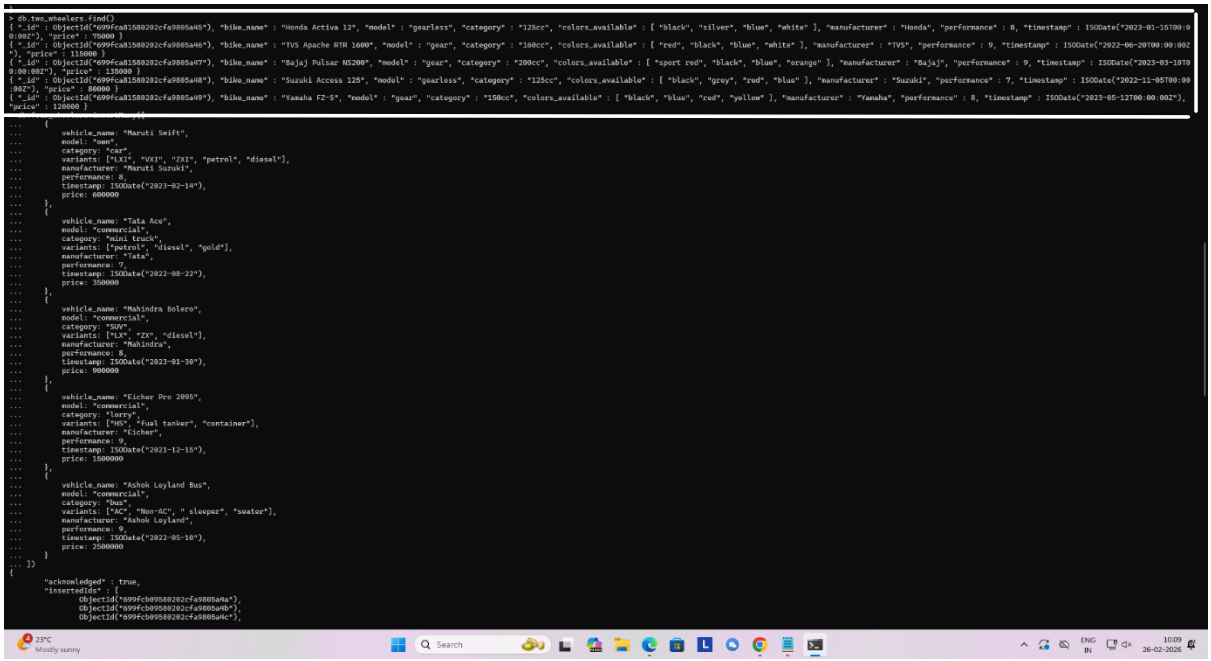


6. Write a MongoDB query to display all documents available in two_wheelers and four_wheelers.



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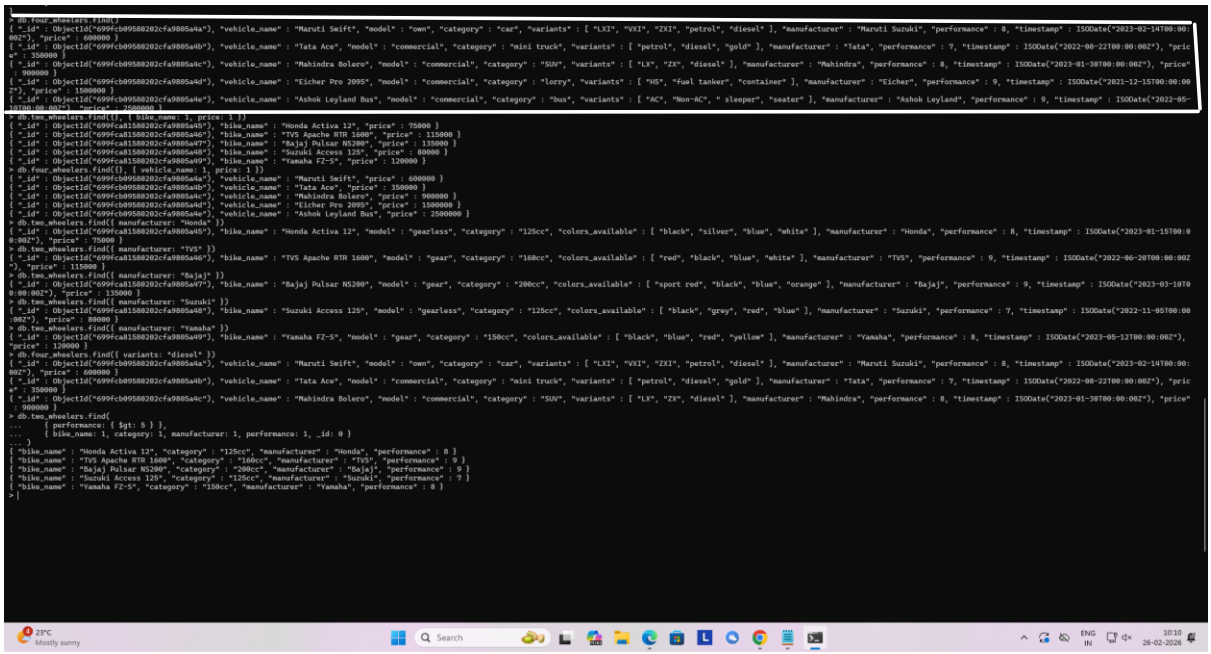
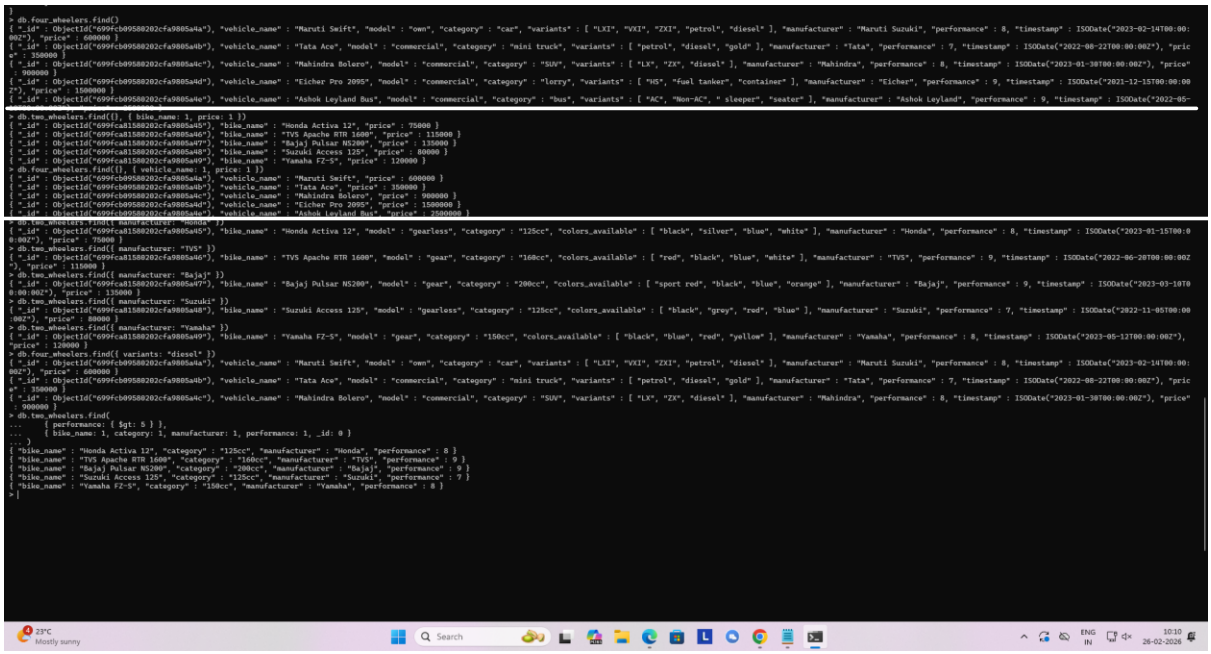


7. Write a MongoDB query to display only vehicle name and price in all the collection of the database



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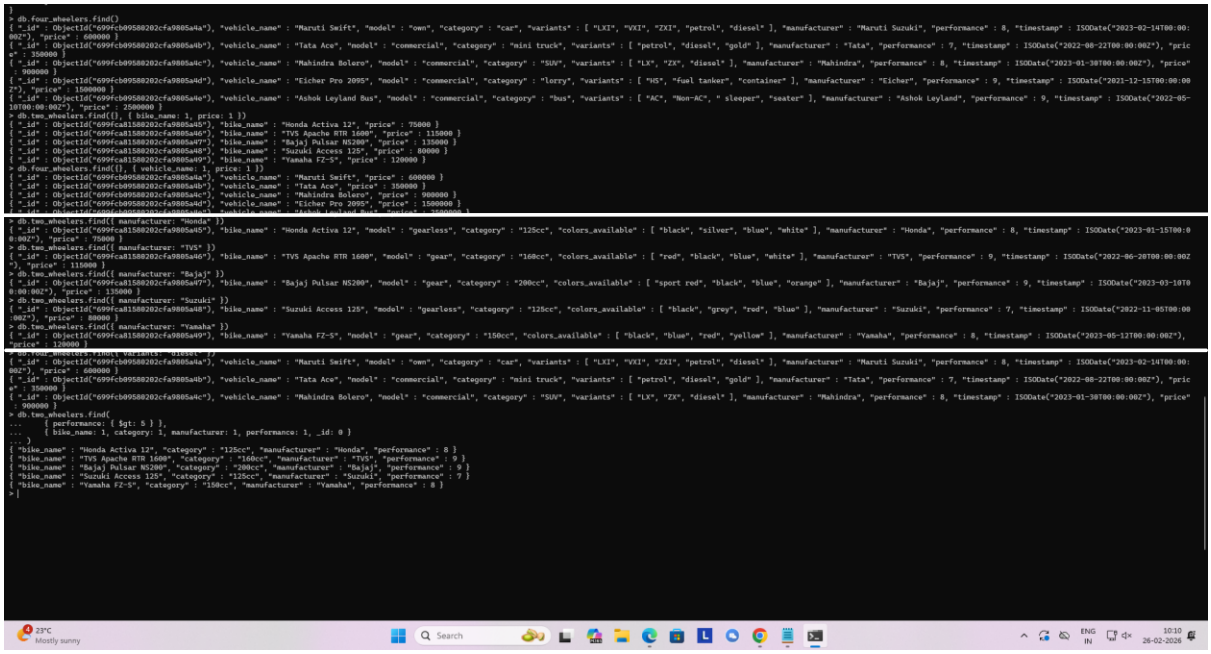
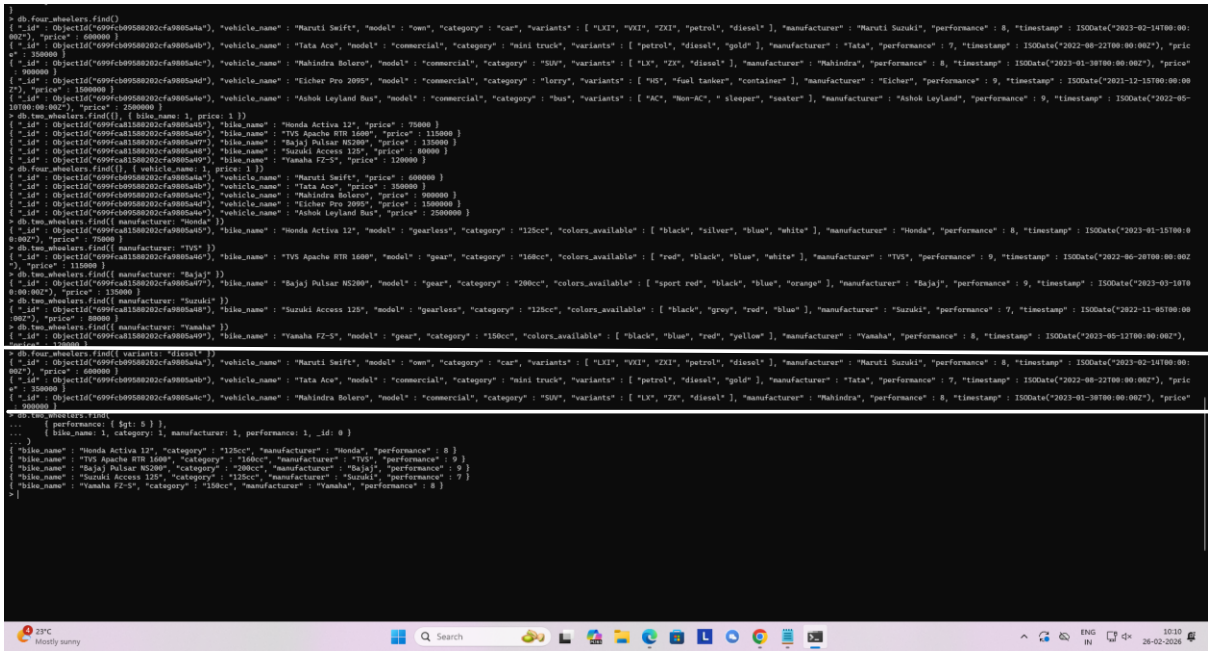
8. Write a MongoDB query to display two_wheelers from a particular company



9. Write a MongoDB query to display four_wheelers available in diesel variants



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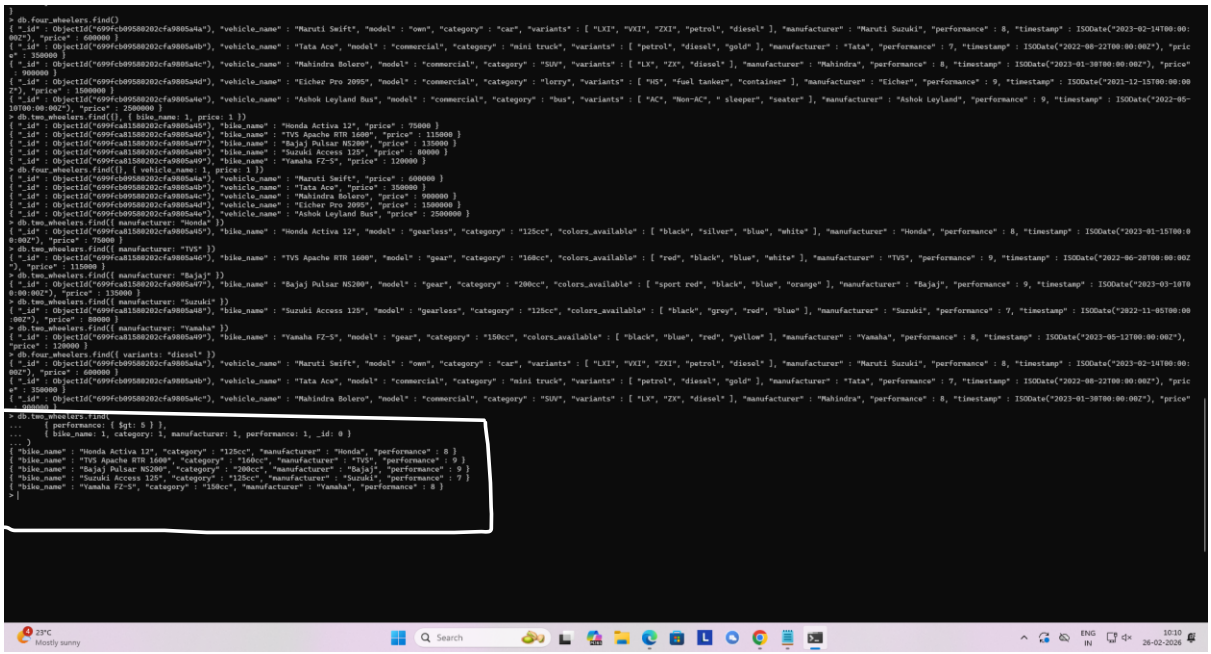
10. Write a MongoDB query to display vehicles name, category and manufacturer details whose rating is more than 5.



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---

2. Use MongoDB to implement the following DB operations for a Zoo

1. Create a database called 'animal' and *write* a MongoDB query to select database as 'animal'.

```
> use animals
switched to db animals
> db
animals
> show dbs
admin     0.000GB
config    0.000GB
local     0.000GB
luffy     0.000GB
school    0.000GB
student   0.000GB
vehicles  0.000GB
> db.createCollection("wild_animals", { capped: true, size: 100000, max: 100 })
{ "ok" : 1 }
> db.createCollection("domestic_animals")
{ "ok" : 1 }
> db.wild_animals.insertMany([
...     {
...         animal_name: "Lion",
```

2. Write a MongoDB query to display all the databases.

```
> use animals
switched to db animals
> db
animals
> show dbs
admin     0.000GB
config    0.000GB
local     0.000GB
luffy     0.000GB
school    0.000GB
student   0.000GB
vehicles  0.000GB
> db.createCollection("wild_animals", { capped: true, size: 100000, max: 100 })
{ "ok" : 1 }
> db.createCollection("domestic_animals")
{ "ok" : 1 }
> db.wild_animals.insertMany([
...     {
...         animal_name: "Lion",
```

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---

3. Create a collection called 'wild_animals'.(use capping) and Create a collection called 'domestic_animals'.

```
> use animals
switched to db animals
> db
animals
> show dbs
admin      0.000GB
config     0.000GB
local      0.000GB
luffy      0.000GB
school     0.000GB
student    0.000GB
vehicles   0.000GB
> db.createCollection("wild_animals", { capped: true, size: 100000, max: 100 })
{ "ok" : 1 }
> db.createCollection("domestic_animals")
{ "ok" : 1 }
> db.wild_animals.insertMany([
...     {
...         animal_name: "Lion",
```

4. Add 5 wild_animal details to the collection named 'wild_animals'. Each document consists of following fields as animal_name, nature (harm or harmless), favorite_foods (meat, rabbits, deer etc) as array, care_taker_name, life span (in years), timestamp (when the animal registered at the Zoo) and expenses.

```
> db.wild_animals.insertMany([
...     {
...         animal_name: "Lion",
...         nature: "harm",
...         favorite_foods: ["meat", "zebras", "deer"],
...         care_taker_name: "John Smith",
...         life_span: 14,
...         timestamp: ISODate("2023-01-15"),
...         expenses: 50000
...     },
...     {
...         animal_name: "Elephant",
...         nature: "harmless",
...         favorite_foods: ["grass", "fruits", "bark", "roots"],
...         care_taker_name: "Rajesh Kumar",
...         life_span: 60,
...         timestamp: ISODate("2022-06-20"),
...         expenses: 80000
...     },
...     {
...         animal_name: "Tiger",
...         nature: "harm",
...         favorite_foods: ["meat", "deer", "wild boar"],
...         care_taker_name: "Mike Johnson",
...         life_span: 16,
...         timestamp: ISODate("2023-03-10"),
...         expenses: 55000
...     },
...     {
...         animal_name: "Giraffe",
...         nature: "harmless",
...         favorite_foods: ["leaves", "acacia", "fruits"],
...         care_taker_name: "Sarah Williams",
...         life_span: 25,
...         timestamp: ISODate("2022-11-05"),
...         expenses: 45000
...     },
...     {
...         animal_name: "Leopard",
...         nature: "harm",
...         favorite_foods: ["meat", "monkeys", "birds"],
...         care_taker_name: "David Brown",
...         life_span: 12,
...         timestamp: ISODate("2023-05-12"),
...         expenses: 40000
...     }
... ])
{
        "acknowledged" : true,
        "insertedIds" : [
                ObjectId("699fcf9c580202cfa9805a4f"),
                ObjectId("699fcf9c580202cfa9805a50"),
                ObjectId("699fcf9c580202cfa9805a51"),
                ObjectId("699fcf9c580202cfa9805a52"),
                ObjectId("699fcf9c580202cfa9805a53")
        ]
}
> db.domestic_animals.insertMany([
...     {
...         animal_name: "Dog",
```

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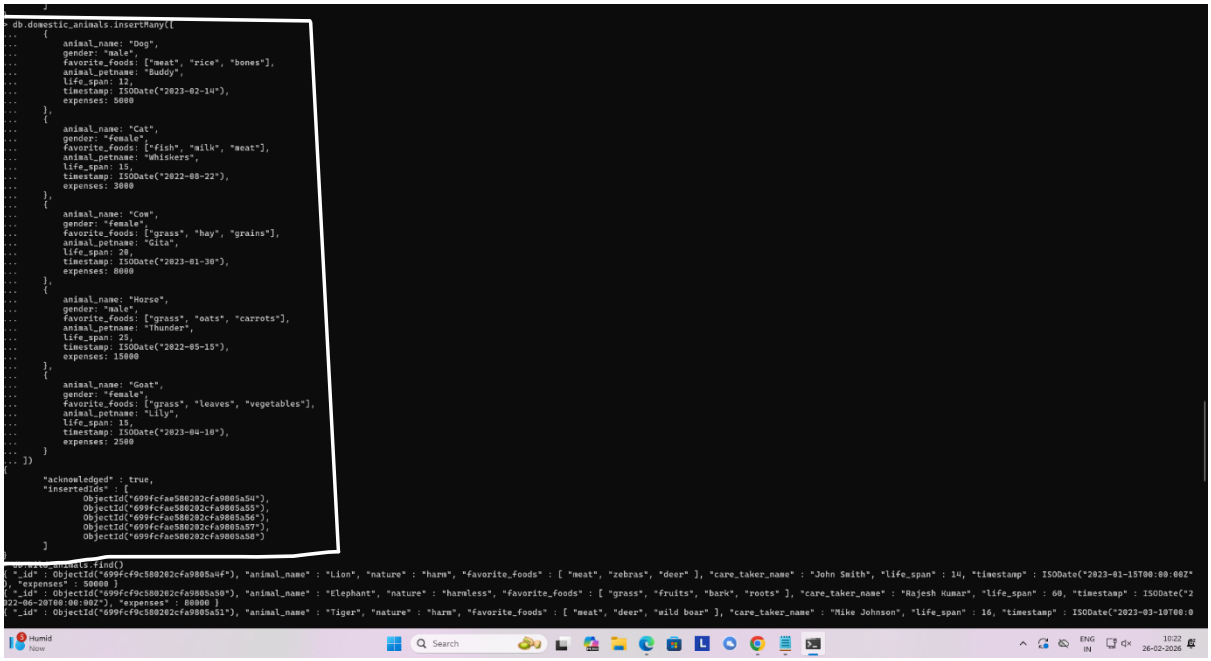
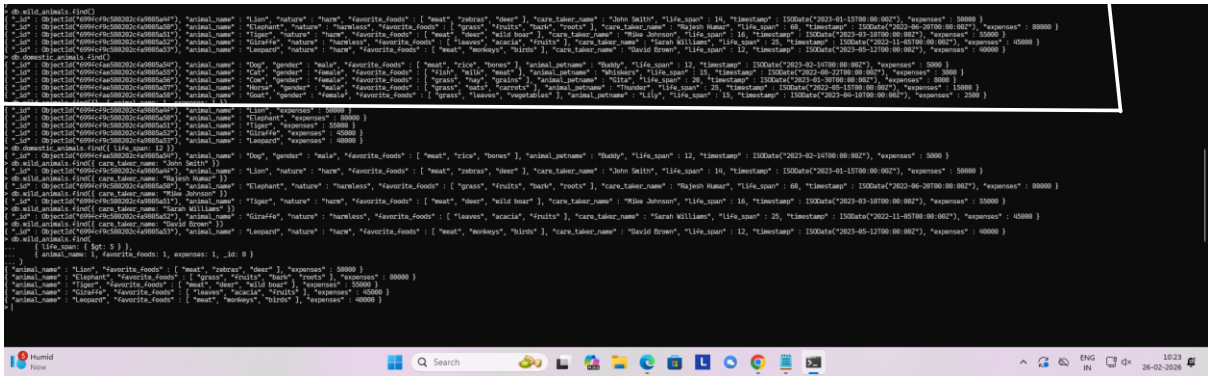
---

5. Add 5 domestic-animal details to the collection named 'domestic_animals'. Each document consists of following fields as animal_name, gender (male or female), favorite_foods (meat, rabbits, deer etc) as array, animal_petname, life span (in years), timestamp (when the animal registered at the Zoo) and expenses.



6. Write a MongoDB query to display all documents available in wild_animals and domastic_animals.



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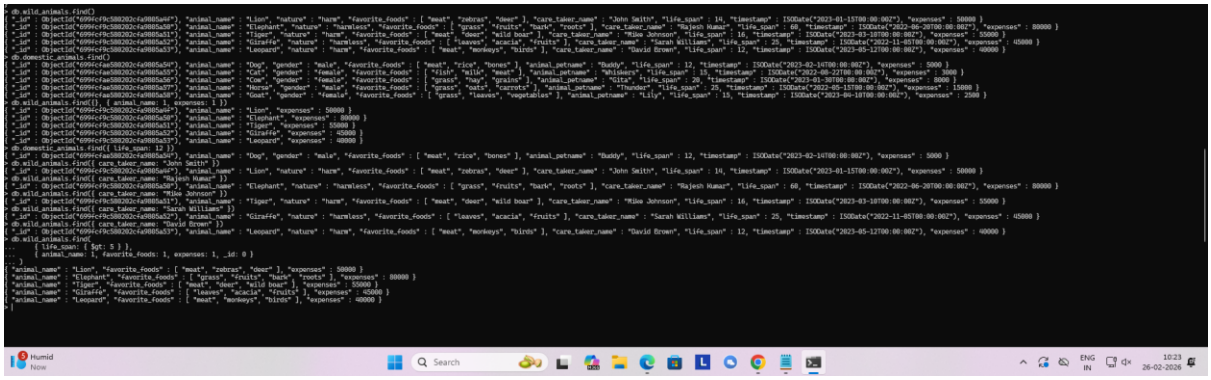
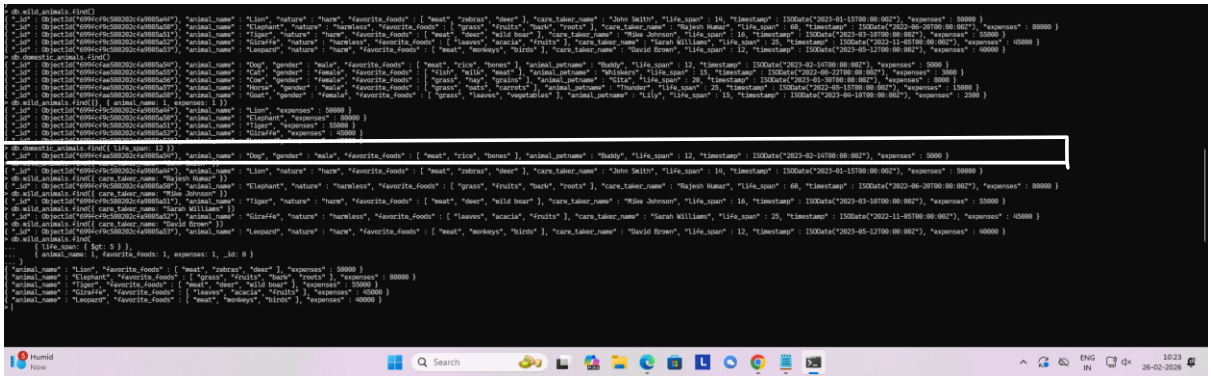
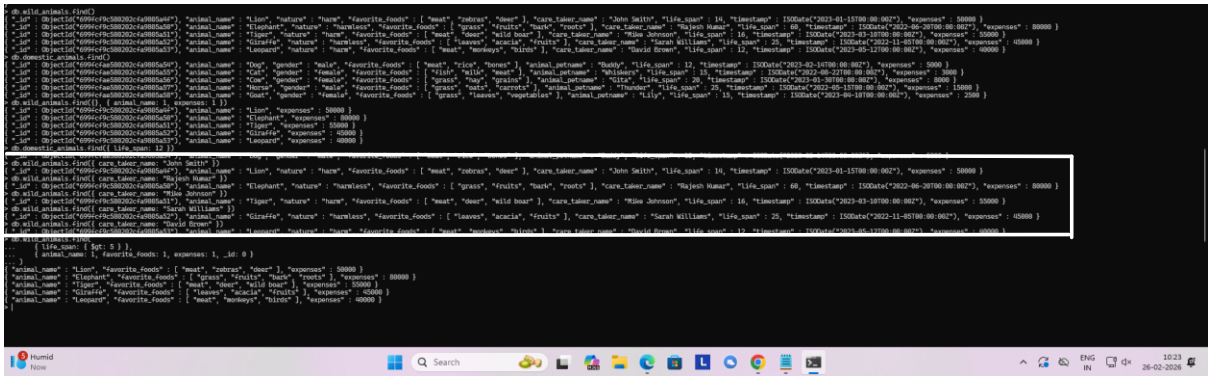
7. Write a MongoDB query to display only animal name and expenses in all the collection of the database



8. Write a MongoDB query to display domestic_animals whose life is a particular year



9. Write a MongoDB query to display wild_animals available under a particular care_taker



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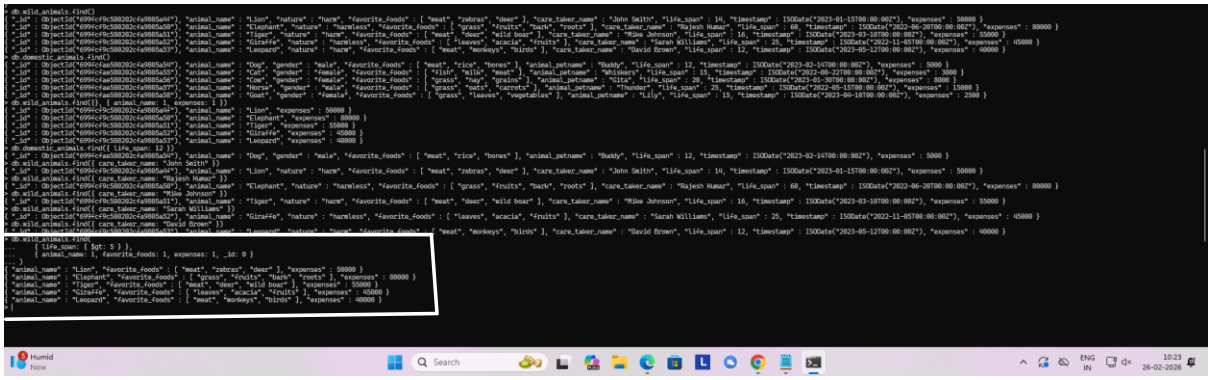
10. Write a MongoDB query to display animal name, favorite_foods and expenses details whose lifespan is more than 5 years.